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RESEARCH

Intent-aware spatiotemporal risk field for multi-AGV path planning with heterogeneous dynamic obstacles

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Abstract

Modern warehouse automation increasingly relies on multi-agent path coordination under heterogeneous, dynamic constraints. Most existing MAPF approaches treat all obstacles uniformly, overlooking the fundamental distinction between communicable agent peers and non-communicable moving obstacles (e.g., forklifts, hand-carts). This paper presents a heterogeneous dynamic obstacle-aware multi-AGV planning framework comprising three integrated components: (1) a Transformer-based trajectory predictor for external obstacles with calibrated uncertainty quantification (87% action accuracy, ADE=1.14); (2) an intent-aware spatiotemporal risk map that differentiates threat levels based on obstacle type and predicted intent; (3) a dual-layer planning architecture integrating risk-aware A* for collision avoidance with an iterative conflict resolution mechanism for multi-agent coordination. Experiments on 13 benchmark maps demonstrate the approach outperforms baselines (PBS, LNS) across 5–10 agent scenarios in success rate, collision count, and makespan. The method achieves 25–370× computational speedup over prediction-enhanced baselines while maintaining superior path quality. Ablation studies validate each module's contribution, demonstrating that coordination mechanisms dominate performance in high-density scenarios.

Keywords Multi-agent path finding · Trajectory prediction · Dynamic obstacle avoidance · Risk-aware planning · Warehouse automation

1 Introduction

In recent years, with the continuous expansion of e-commerce and the rapid iteration of intelligent logistics technologies, warehouse automation has become a core pillar of modern logistics systems (Boysen et al. 2019). An increasing number of large-scale warehouse facilities have begun deploying multi-Automated Guided Vehicle (AGV) systems to enable automated goods transportation and sorting (Azadeh et al. 2017). For instance, Amazon's Kiva robot system coordinates thousands of mobile robots across multiple facilities globally (Wurman et al. 2008), while JD.com operates fully unmanned warehouses deploying hundreds of AGVs in each facility to achieve highly automated order processing workflows. In such systems, multiple AGVs must navigate concurrently within the same warehouse environment, each moving from a designated start location to a designated goal while avoiding collisions with both peer AGVs and environmental obstacles. How to efficiently and safely compute collision-free paths for multiple AGVs has become a critical challenge in warehouse automation (Yang et al. 2026).

The core challenge of multi-AGV path planning lies in coordinating multiple agents' movements to avoid inter-agent collisions while handling dynamic obstacles. This problem builds upon the well-studied Multi-Agent Path Finding (MAPF) framework (Stern et al. 2019). Classical MAPF algorithms have made significant progress: the Conflict-Based Search (CBS) (Sharon et al. 2015) uses two-level search for conflict resolution; the Large Neighborhood Search (LNS) family (Li et al. 2021a, 2022) employs destroy-and-repair strategies with excellent scalability; and Priority-Based Search (PBS) (Ma et al. 2019) achieves efficiency through priority ordering. These methods have been successfully applied to warehouse scheduling (Li et al. 2021b), traffic management (Ho et al. 2019), and robot coordination (Hönig et al. 2016).

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However, a critical limitation of these algorithms is their assumption of static environments.Real warehouses contain numerous dynamic obstacles, including manually operated forklifts, hand carts, and human workers whose trajectories can only be inferred through observation (González et al. 2015). Most existing MAPF algorithms are not designed to handle such dynamic external obstacles. When dynamic obstacles are present, paths computed by static MAPF algorithms may face collision risks during execution, failing to guarantee safety. Reactive single-agent methods like the Dynamic Window Approach (DWA) (Fox et al. 2002) and Artificial Potential Field (APF) (Khatib 1986) can handle dynamic obstacles but lack multi-agent coordination mechanisms, resulting in inter-agent conflicts in multi-AGV scenarios (Koren et al. 1991). Some attempts have been made to extend MAPF to dynamic environments (Ma et al. 2018), but most rely on reactive strategies that lack predictive foresight, leading to inefficient motion and frequent stops.

Furthermore, a critical overlooked aspect is the fundamental heterogeneity of warehouse obstacles. Traditional MAPF assumes uniform obstacle handling, but real warehouses contain two distinct types: (1) communicable AGV peers with predictable, negotiable behavior; and (2) non-communicable external obstacles (forklifts, personnel) with behavior inferred only from observations. These differ critically in information availability, predictability, and controllability (Luo et al. 2020). Existing work (Ma et al. 2018; Liu et al. 2020),neglects this distinction, limiting effectiveness in realistic warehouse environments.

This paper proposes a heterogeneous dynamic obstacle-aware multi-AGV planning framework addressing the above challenges. The main contributions are: (1) explicit classification and differentiated handling of communicable agents versus non-communicable obstacles, with distinct avoidance strategies tailored to their heterogeneous characteristics; (2) Transformer-based trajectory prediction with calibrated uncertainty quantification, achieving 87% action accuracy with low prediction error (ADE=1.14, FDE=1.78), enabling probabilistic risk assessment for external obstacles; (3) integrated dual-layer planning architecture combining risk-aware A* search with iterative conflict resolution, preserving MAPF coordination mechanisms while handling dynamic uncertainty. Extensive experiments on 13 benchmark maps demonstrate significant improvements over existing methods in success rates, collision avoidance, and computational efficiency.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the formal problem formulation. Section 4 provides a detailed description of the proposed method. Section 5 presents experimental results and analysis. Section 6 discusses limitations and future directions. Section 7 concludes the paper.

2 Related work

Multi-Agent Path Finding (MAPF) has been an active research area in robotics and artificial intelligence for decades (Stern et al. 2019). Among search-based methods, CBS (Sharon et al. 2015) adopts a two-level search structure, where the low level independently plans optimal paths for each agent and the high level detects spatiotemporal conflicts among agents and generates constraint branches for conflict resolution. Barer et al. (2014) proposed ECBS, which accelerates the expansion of high-level nodes through bounded suboptimal search. Li et al. (2021a, 2022) proposed LNS and LNS2, which adopt a Large Neighborhood Search framework and demonstrate excellent scalability and solution quality in large-scale scenarios. Ma et al. (2019) proposed PBS, which avoids conflicts through priority ordering and features low computational complexity. Additionally, the subdimensional expansion method proposed by Wagner and Choset (2015) and the integer linear programming method proposed by Surynek et al. (2016) have provided alternative solving perspectives for the MAPF problem. Recently, deep reinforcement learning has also been applied to MAPF (Sartoretti et al. 2019; Long et al. 2018); however, its generalization capability in large-scale scenarios remains limited. These methods have been successfully applied to warehouse scheduling (Li et al. 2021b), traffic management (Ho et al. 2019), and robot coordination (Hönig et al. 2016).

A critical limitation of these classical MAPF algorithms is their fundamental assumption that obstacles in the environment are static. Notably, most MAPF algorithms (Sharon et al. 2015; Li et al. 2021a; Ma et al. 2019) are not designed to directly handle scenarios involving dynamic external obstacles.

Dynamic obstacle planning has been extensively studied in single-agent and multi-agent contexts. Reactive single-agent methods represent an early class of solutions. The Dynamic Window Approach (DWA) proposed by Fox et al. (2002) searches for optimal velocity commands in velocity space, balancing obstacle avoidance and goal-directed behavior; however, it lacks a global perspective and is prone to local optima (Brock and Khatib 1999). The Artificial Potential Field (APF) method proposed by Khatib (1986) guides robot motion through attractive and repulsive force fields; while conceptually simple, it suffers from the local minimum problem (Ulrich and Borenstein 1998). The Vector Field Histogram (VFH) method proposed by Ulrich and Borenstein (Tai et al. 2017) has also been widely applied in two-dimensional dynamic obstacle avoidance. These reactive methods can handle dynamic obstacles but are primarily designed for single agents, lacking coordination mechanisms among multiple agents.

Complementary to reactive approaches, temporally-aware planning methods have been developed to handle dynamic

obstacles more explicitly. Safe Interval Path Planning (SIPP), proposed by Phillips and Likhachev (2011), extends the state space by adding temporal dimensions to create safe time intervals where agents can safely occupy grid positions. This approach enables precise spatiotemporal reasoning and can guarantee collision-free paths when obstacle trajectories are known. Reservation-table methods and time-expanded planning graph approaches similarly leverage explicit temporal reasoning, allowing agents to reserve spatiotemporal nodes in advance and avoid future conflicts through forward-looking planning (Silver 2005). Velocity Obstacle (VO) methods, including Reciprocal Collision Avoidance (RCA) and Reciprocal Velocity Obstacles (RVO), provide a geometric framework for computing avoidance maneuvers by formulating collision constraints in velocity space (Berg et al. 2008). These methods are effective in continuous spaces but require adaptation to discrete grid-based MAPF environments. More critically, they assume known obstacle trajectories (SIPP) or reciprocal cooperation among agents (RVO), assumptions that do not hold for non-communicable external obstacles in warehouse environments.

Predictive methods represent a more sophisticated approach, planning avoidance paths in advance by forecasting future obstacle trajectories to achieve superior foresight and path efficiency. Commonly used classical prediction methods include Kalman filtering (Ahmad and Namerikawa 2013), Gaussian process regression (Rong et al. 2019), and kinematic models such as constant velocity and constant acceleration (Schöller et al. 2020). When prediction uncertainty is unavoidable, chance-constrained planning approaches provide a principled framework to reason about probabilistic obstacles and guarantee collision avoidance with specified confidence levels (Blackmore et al. 2011). However, these classical predictive methods are predominantly designed for individual agents and have limited integration with multi-agent coordination frameworks, limiting their applicability in multi-AGV scenarios.

In trajectory prediction, early deep learning methods leveraged recurrent neural networks for sequential modeling: Alahi et al. (2016) proposed Social LSTM, which captures temporal features of trajectories through RNNs; Gupta et al. (2018) proposed Social GAN, which introduces a generative adversarial framework to enhance prediction diversity. These methods have limited capability in capturing complex motion patterns compared to modern architectures.

More recently, Transformer-based trajectory prediction methods have achieved remarkable progress in pedestrian trajectory prediction (Giuliari et al. 2021; Yu et al. 2020) and vehicle trajectory prediction (Ngiam et al. 2021). Transformers offer advantages in parallel processing and interpretability through attention mechanisms for long-range interactions. To quantify prediction uncertainty, probabilistic output frameworks have been introduced (Makansi et al. 2019), enabling models to simultaneously output the mean and variance of predicted trajectories. This uncertainty quantification is particularly important for safety-critical path planning tasks. However, existing trajectory prediction studies typically treat prediction and intent classification as two separate tasks, lacking a unified framework (Rashid and Yusof 2026). Recent advances in 2023-2024 have focused on integrating vision-language models for intent understanding and graph neural networks for structured environment representation (Mishra et al. 2026), though applications to warehouse-scale multi-agent systems remain limited.

The present work lies at the intersection of MAPF and dynamic path planning. The distinctions from existing work can be articulated from four dimensions. First, compared with static MAPF algorithms, the proposed method handles dynamic external obstacles, enabling the MAPF framework to operate safely in dynamic environments. Second, compared with single-agent dynamic path planning methods, the proposed method addresses coordination among multiple agents, avoiding path conflicts and deadlocks. Third, compared with simple reactive dynamic handling approaches, the proposed method introduces Transformer-based trajectory prediction and intent awareness, rendering the avoidance strategy forward-looking and targeted. Fourth, compared with studies that treat all obstacles homogeneously, the proposed method distinguishes the heterogeneous characteristics of communicable and non-communicable obstacles and adopts differentiated avoidance strategies.

3 Problem formulation

3.1 Environment model

As illustrated in Fig. 1, the warehouse environment is modeled as a two-dimensional grid map $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where$\mathcal{V}$ denotes the set of grid nodes, each node$v \in \mathcal{V}$ corresponding to a coordinate(x, y) on the map, and $\mathcal{E}$ denotes the set of connected edges between adjacent nodes. The grid nodes are classified into two categories:

1.Free space $\mathcal{V}_{free}$: regions where agents can move freely;

2.Obstacle space $\mathcal{V}_{obs}$: static impassable structures including walls, shelving units, and other fixed obstacles.

At each discrete time step $t \in \{0, 1, \ldots, T\}$, an agent can move to one of its four neighboring nodes or remain at its current position, i.e., the action space is $a \in \{(0, 0), (\pm 1, 0), (0, \pm 1)\}$.

3.2 Agent and heterogeneous obstacles

Each communicable agent $a_i \in A$ maintains a local information set $I_i(t) = \{p_i(t), \pi_i, \rho_i\}$, where $p_i(t)$ is the current position, π_i is the planned spatiotemporal path, and $\rho_i \in \mathbb{N}$

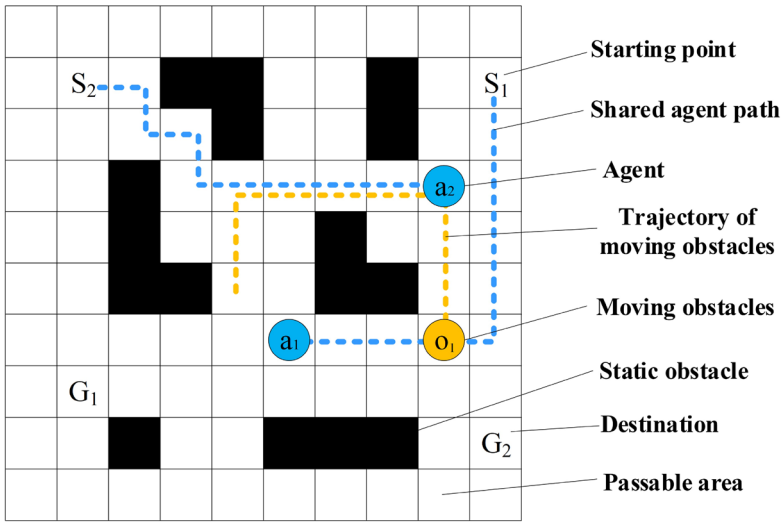


Fig. 1 Multi-agent warehouse environment with heterogeneous obstacles: agents (blue), static obstacles (black regions), and dynamic obstacles (yellow)

denotes the agent's priority rank. Agents broadcast $I_i(t)$ at every time step over a shared communication channel; these broadcasts are aggregated by a central coordination module that performs global conflict detection and drives the iterative conflict-resolution procedure (see Algorithm 1). Centralized detection avoids inconsistencies that can arise from purely local conflict checks (for example, symmetric simultaneous yielding that can produce deadlock) and ensures a unique 'IdentifyReplanAgent(c)' decision for each detected conflict. When a spatiotemporal conflict is detected between agents a_i and a_j, the agent with the larger numerical priority value (i.e., lower precedence) is designated as the yielding agent and must either replan its path or insert a wait action. Priority ordering is established deterministically according to agent identifiers, with lower identifier values indicating higher precedence; this assignment remains fixed during a single planning epoch to ensure deterministic conflict resolution and reproducibility.

Consider a multi-agent system comprising n homogeneous agents $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$. Each agent a_i is assigned a start $s_i \in \mathcal{V}_{\text{free}}$ and a goal $g_i \in \mathcal{V}_{\text{free}}$. The environment also contains m non-communicable external dynamic obstacles $\mathcal{O} = \{o_1, o_2, \ldots, o_m\}$ (e.g., forklifts, personnel).

Agent state. The position of agent a_i at time t is $p_i(t) = (x_i, y_i) \in \mathcal{V}_{\text{free}}$. Its complete path is:

$$\pi_i = \{p_i(0), p_i(1), \ldots, p_i(T_i)\} \tag{1}$$

where $p_i(0) = s_i$, and there exists a time step T_i such that $p_i(T_i) = g_i$.

External obstacle state. The position of obstacle o_k at time t is denoted as $p_k^{ext}(t) = (x_k^{ext}, y_k^{ext})$. Due to its unpredictable and non-communicable behavior, its position sequence is generally uncertain from the system's perspective.

3.3 Collision types

To ensure path safety, the system must satisfy the following four collision constraints:

1.Agent-to-agent vertex collision: No two agents may occupy the same grid cell at the same time step.

$$\forall i \neq j, \forall t : p_i(t) \neq p_j(t) \tag{2}$$

2.Agent-to-agent edge collision: No two agents may swap positions across adjacent time steps (i.e., passing through each other).

$$\forall i \neq j, \forall t : \neg\Big(p_i(t) = p_j(t{+}1) \wedge p_i(t{+}1) = p_j(t)\Big) \tag{3}$$

3.Obstacle vertex collision: The position of any agent at time t must not coincide with the position of any external obstacle.

$$\forall i, \forall k, \forall t : p_i(t) \neq p_k^{ext}(t) \tag{4}$$

4.Obstacle edge collision: No agent may swap positions with any external obstacle across adjacent time steps.

$$\forall i, \forall k, \forall t : (p_i(t) = p_k^{ext}(t{+}1)) \wedge (p_i(t{+}1) = p_k^{ext}(t)) = \varnothing \tag{5}$$

3.4 Constraints and optimization objective

1. **Kinematic constraints**: Each agent's path satisfies $\|p_i(t+1) - p_i(t)\|_1 \leq 1$ with $p_i(t) \in \mathcal{V}_{\text{free}}$.

2. **Optimization objective**: We minimize the sum-of-costs $\sum_i f(\pi_i)$, where $f(\pi_i)$ denotes the path length, subject to collision-avoidance constraints with dynamic obstacles and inter-agent coordination requirements. Although

Table 1 Comparison between classical MAPF and the problem addressed in this work

Dimension	Classical MAPF	This work
Environment	Static	Dynamic
Agents	Homogeneous	Heterogeneous
Obstacle prediction	Not required	Required
Risk modeling	Not required	Required
Coordination mechanism	Single-layer	Dual-layer

makespan $\max_i T_i$ is not an explicit optimization objective in our formulation, reductions in collisions and unnecessary waiting commonly yield shorter agent completion times; accordingly, we report makespan as a secondary evaluation metric to capture such indirect improvements (Table 1).

3.5 Distinction from classical MAPF

Key differences from standard MAPF:

4 Proposed method

4.1 System overview

To address the challenge posed by the coexistence of heterogeneous obstacles in warehouse logistics environments, this paper proposes an integrated path planning framework that combines "trajectory prediction – risk-aware planning – multi-agent coordination." The core design philosophy of this framework is a divide-and-conquer strategy for heterogeneous obstacles, which decouples the obstacle handling process into two distinct conflict avoidance mechanisms based on the communication capability and controllability of mobile entities:

1.For non-communicable external obstacles (e.g., forklifts, pedestrians): A "prediction + avoidance" proactive obstacle avoidance mechanism is adopted. The system first employs a Transformer to extract historical motion features and predict future trajectories. Subsequently, the prediction uncertainty is transformed into a Spatiotemporal Risk Map, enabling agents to perceive and circumvent high-risk regions through the cost function during the path search phase.

2. For communicable multi-agent peers: A "coordination + iterative replanning" collaborative mechanism is employed. After generating preliminary avoidance paths, the system applies conflict detection and local replanning operations to resolve inter-agent collisions through priority-based negotiation and spatiotemporal conflict analysis.

The overall execution flow of the system is illustrated in Fig. 2. At each discrete time step t, the system sequentially executes four core modules: (I) Trajectory Prediction Module — computes multimodal predictions of external obstacle trajectories with uncertainty quantification; (II) Risk Field Generation Module — transforms predictions into spatiotemporal risk maps with intent-aware weighting; (III) Path Planning Module — generates initial feasible paths using risk-aware A* with Manhattan-distance heuristics; (IV) Conflict Resolution Module — detects and resolves inter-agent collisions through dynamic priority adjustment and localized replanning.

This section presents the integrated framework combining trajectory prediction, risk mapping, and risk-aware planning. The system operates in a four-stage pipeline at each planning step: (I) predict non-communicable obstacle motion; (II) transform predictions into spatiotemporal risk fields; (III) compute risk-aware paths; (IV) resolve inter-agent conflicts.

4.2 Trajectory prediction for external obstacles

Non-communicable obstacles in warehouses exhibit discrete, gridworld-constrained motion. Rather than regressing absolute coordinates, we predict action probability distributions to quantify uncertainty. This probabilistic approach naturally feeds into risk field construction.

4.2.1 Prediction formulation

Given an external obstacle o_k's historical trajectory $\mathcal{H}_k = \{p_k(t - L_{\text{in}} + 1), \ldots, p_k(t)\}$ (10 steps), the model predicts action distributions for the next $L_{\text{out}} = 5$ steps:

$$P(a_{t+\tau} \mid \mathcal{H}_k, a_{t+1:t+\tau-1}), \quad a \in \mathcal{S}_A \tag{6}$$

where the action space $\mathcal{S}_A = \{\text{stay, up, down, left, right}\}$ corresponds to the discrete displacements in the grid map. The future position $\hat{p}_{t+\tau}$ is computed through the recursive position update formula:

$$\hat{p}_{t+\tau} = \hat{p}_{t+\tau-1} + \Delta(a_{t+\tau}) \tag{7}$$

where $\Delta(\cdot)$ denotes the mapping from actions to grid displacements. Correspondingly, the prediction uncertainty $u_{t+\tau}$ at each time step is derived from the maximum confidence of the classification probability:

$$u_{t+\tau} = 0.5 + 1.5 \cdot (1 - \max P(a_{t+\tau})) \tag{8}$$

The constants 0.5 and 1.5 map confidence scores to uncertainty magnitudes $[0.5, 2.0]$. When $\max P = 1$ (perfect prediction), $u = 0.5$ provides baseline diffusion; when $\max P = 0.2$ (uniform distribution over the five-action vocabulary), $u = 1.7$ yields substantially broader diffusion. This calibration ensures risk-field proportionality to prediction quality while avoiding destabilization. The lower bound

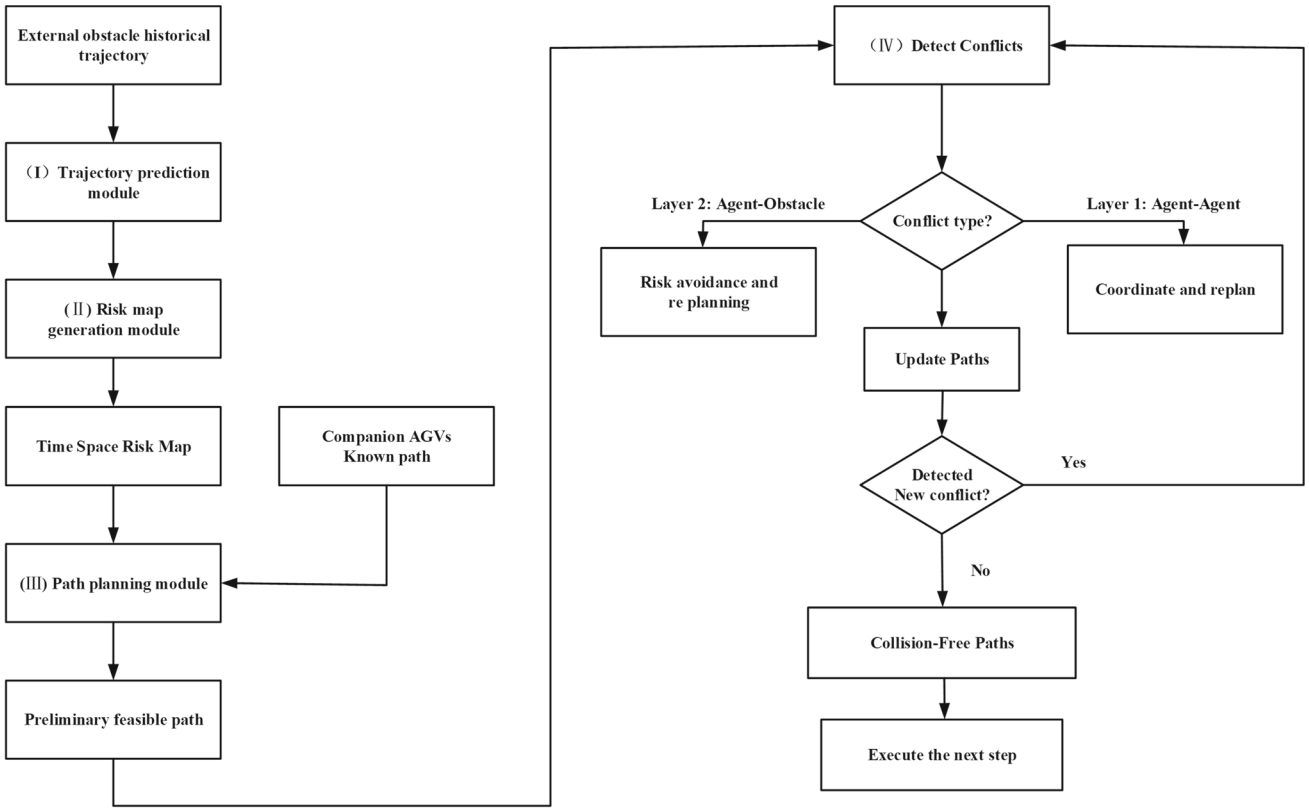


Fig. 2 System pipeline: (**I**) trajectory prediction, (**II**) intent-aware risk map generation, (**III**) risk-aware A* planning, (**IV**) iterative conflict resolution

prevents degenerate kernels, while the upper bound empirically balances precaution with computational tractability.

4.2.2 Network architecture

Figure 3 illustrates the trajectory prediction network, which employs a hybrid encoder-decoder design:

Encoder: Trajectory inputs undergo linear embedding and sinusoidal positional encoding, then pass through a 2-layer Transformer Encoder (4 attention heads, 128-dim FFN per layer). This captures long-range temporal dependencies in obstacle motion histories.

Decoder: A GRUCell with cross-attention performs autoregressive decoding. At each step, the decoder receives encoder context vectors, current action embeddings, and time-horizon information, enabling multi-step future action prediction.

Output Head: A dual-head output structure is adopted, which separately projects and generates the logits sequence of actions and the uncertainty feature sequence.

Architecture Justification: The Transformer-GRU hybrid architecture was selected over alternatives (LSTM, pure GRU, constant-velocity models) based on empirical performance on the held-out split of the 21-map training collection. Compared to constant-velocity models that assume linear motion, the Transformer encoder effectively captures nonlinear turning patterns and corridor-specific behaviors in warehouse obstacles. Compared to pure LSTM or GRU baselines, the Transformer's multi-head attention mechanism better captures long-range spatial correlations in obstacle trajectories (e.g., understanding that an obstacle near an intersection is likely to turn). The GRU decoder balances this global context with computational efficiency during autoregressive generation. Ablation results on the held-out split show that the Transformer encoder achieves 3% higher action prediction accuracy (87% vs 84% for GRU-only), while maintaining comparable computational cost ($\approx$ 12 ms per prediction).

4.2.3 Loss function

The loss combines action classification and trajectory regularization:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + 0.2 \cdot \mathcal{L}_{\text{pos}} \tag{9}$$

$\mathcal{L}_{\text{CE}}$ is standard cross-entropy for action prediction. $\mathcal{L}_{\text{pos}}$ is SmoothL1 regularization on cumulative predicted displacements, reducing drift in multi-step unrolling. The coefficient 0.2 balances primary (action accuracy) and secondary (spa-

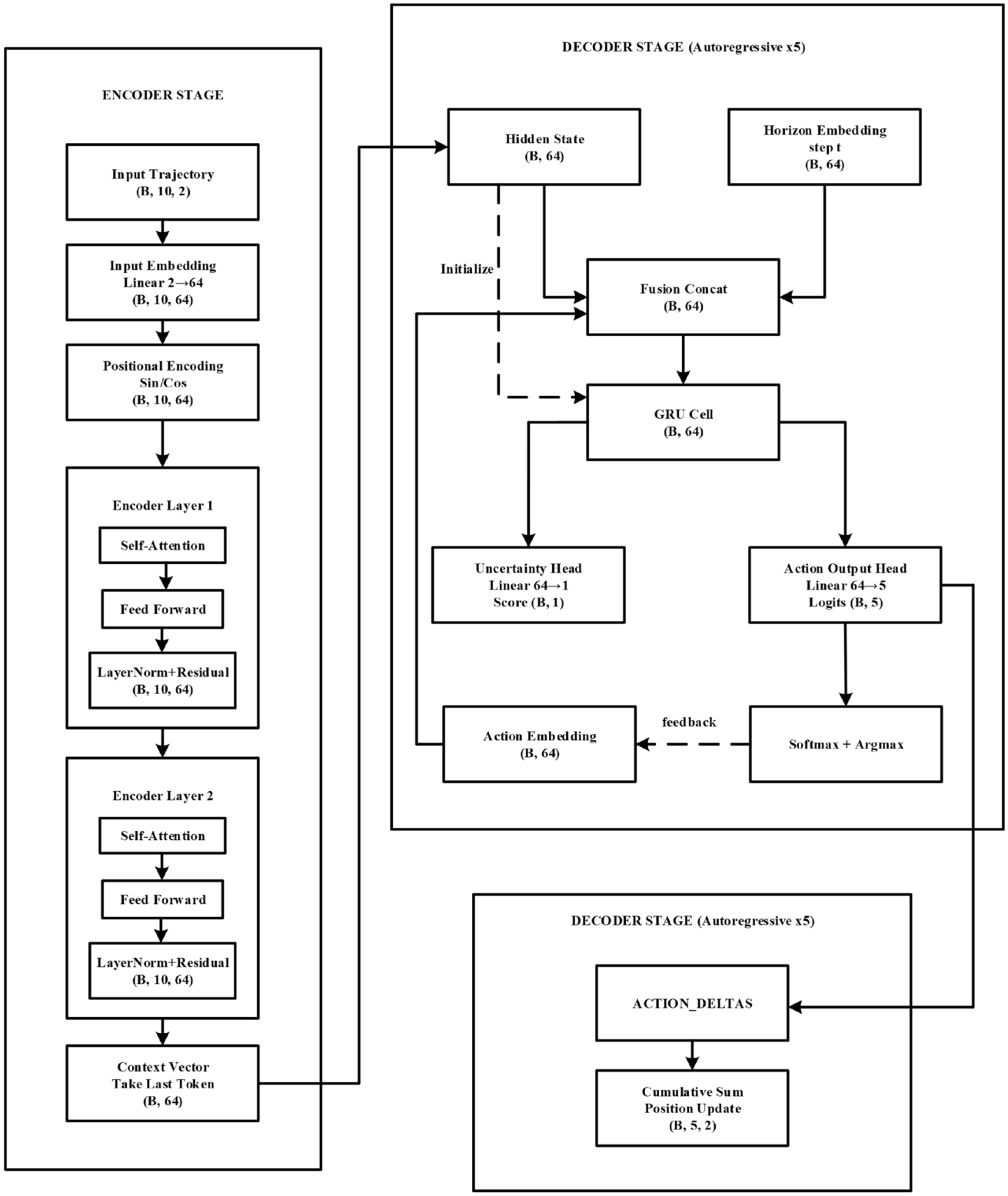


Fig. 3 Architecture of the Transformer-based trajectory prediction network for external obstacles

tial coherence) objectives, and was selected empirically on the held-out split of the 21-map training collection.

4.2.4 Training data and procedure

The predictor was trained on approximately 20,000 obstacle trajectories collected from 21 maps specifically generated for training purposes, which do not overlap with the 13 benchmark maps used in performance evaluation. Each trajectory sequence contains ≥ 200 time steps, providing rich temporal variation across various map structures and obstacle configurations. Training/test split: 80/20. Optimization: Adam (lr=0.001, batch size=64, 80 epochs, early stopping on test loss). The substantial training data volume and multi-map diversity ensure robust model generalization across warehouse scenarios.

Test-set quantitative results (Approximately 4,000 samples from the 20% holdout split of the 21-map training collection): (i) Action accuracy: 87%, validating that the 5-action vocabulary captures obstacle behavior; (ii) ADE: 1.14 $\pm$ 0.08 grid cells, demonstrating tight trajectory tracking; (iii) FDE: 1.78 $\pm$ 0.35 grid cells (5-step horizon), showing acceptable long-horizon drift; (iv) Position error: 0.42 grid cells, confirming spatial consistency. These metrics confirm adequate prediction quality for downstream risk mapping and planning.

4.3 Intent-aware spatiotemporal risk map

After obtaining the action probability distribution of external obstacles, the system needs to transform it into costs that can be directly utilized by the path planner. This section constructs a dynamic risk map $\mathcal{R}$ by combining discrete intent inference with Gaussian diffusion kernels.

4.3.1 Binary intent inference

In grid environments, obstacle motion reduces to straight movement (no direction change) and turning (direction change). The intent is determined from displacement continuity: if $\Delta p_t = \Delta p_{t-1}$, intent is "straight"; otherwise "turning." This binary classification reflects kinetic threat: straight-moving obstacles maintain momentum, while turning obstacles may decelerate or stop.

The composite risk coefficient combines intent and obstacle type: $\eta_k = w(\text{intent}_k) \cdot c_k$, where w is intent weight and $c_k = \max P(a_t)$ is action confidence. Weight assignment: $w_{\text{straight}} = 0.35$, $w_{\text{turning}} = 0.2$. This reflects warehouse physics: straight-ahead obstacles (forklifts on corridors) maintain higher momentum and longer braking distances, posing greater kinetic threat despite lower directional uncertainty. Turning obstacles have reduced momentum, justifying lower weights despite higher directional variability.

4.3.2 Risk field generation

Predicted positions are extended into probabilistic kernels. Risk at node v at time t combines three components:

$$G_k(v,t) = \exp\left(-\frac{\|v - \hat{\mu}_k(t)\|^2}{2\sigma_k(t)^2}\right), \quad \sigma_k(t) = \max(u_k(t), 0.5) \tag{10}$$

2.Time decay factor: A decay function $\alpha(t)$ is introduced to reflect the decreasing reliability of predictions over longer time horizons:

$$\alpha(t) = \frac{1}{1 + 0.8t} \tag{11}$$

3. Individual obstacle risk: By combining the intent coefficient with spatiotemporal decay, the risk value generated by a single obstacle o_k is formulated as:

$$R_k(v,t) = \eta_k \cdot \alpha(t) \cdot G_k(v,t) \tag{12}$$

Global risk aggregation: The final risk value $\mathcal{R}(v,t)$ at node v is taken as the maximum over all dynamic obstacles. Meanwhile, to unify the cost function, the risk value of static obstacles $\mathcal{V}_{obs}$ is assigned the maximum weight:

$$\mathcal{R}(v,t) = \begin{cases} 1.0, & \text{if } v \in \mathcal{V}_{obs} \\ \max_k R_k(v,t), & \text{otherwise} \end{cases} \tag{13}$$

Through this modeling approach, the originally isolated predicted actions are transformed into a spatiotemporally continuous pressure field, enabling the underlying planner to anticipate and circumvent high-risk regions.

Static obstacles $v \in \mathcal{V}_{obs}$ are treated as hard constraints in the path planner: they are permanently removed from the free space $\mathcal{V}_{free}$ and are therefore non-traversable. The assignment $R(v,t) = 1.0$ is introduced solely to unify the formulation of the cost function. In practice, the Risk-aware A* algorithm directly rejects any node $v \in \mathcal{V}_{obs}$ during node expansion, regardless of the value of λ.

4.4 Risk-aware path planning and coordination

The spatiotemporal risk map feeds into a two-level planning framework: individual risk-aware A* searches compute initial paths; iterative conflict resolution coordinates multi-agent conflicts. This divide-and-conquer approach manages both dynamic obstacles and agent-agent coordination.

4.4.1 Risk-aware A* algorithm

Standard A* is augmented to incorporate risk awareness. The cost function for node v at time τ becomes:

$$f(v,\tau) = g(v,\tau) + h(v) + \lambda \cdot \mathcal{R}(v,\tau) \tag{14}$$

where g = accumulated path cost; $h(v)$ = Manhattan heuristic; $\mathcal{R}(v,\tau)$ = risk value; λ = risk weight (balance between efficiency and safety), Setting $\lambda = 1.0$ provides equal weighting; adjusting λ prioritizes risk avoidance or speed.

By incorporating $\lambda \cdot \mathcal{R}(v,\tau)$, the search algorithm tends to select grid cells with lower risk values. If the risk value of a candidate node approaches infinity (e.g., static obstacles or regions with extremely high collision risk), that node is deemed impassable.

4.4.2 Collaborative planning framework

To handle spatiotemporal conflicts among agent peers (i.e., vertex collisions and edge collisions), the system adopts an iterative conflict resolution (ICR) mechanism inspired by neighborhood search paradigms. Unlike classical LNS that employs explicit neighborhood destruction and repair operators, the proposed ICR operates in a sliding-window fashion: it detects conflicts among all agents and sequentially resolves them through localized replanning or wait-action insertion, which is more suitable for real-time dynamic environments. Its core logic is presented in Algorithm 1.

4.4.3 Differentiated handling analysis

This framework embodies the divide-and-conquer design of this work:

1.Continuous risk handling: The prediction uncertainty of external obstacles is quantified as $Risk_k(v,\tau)$, and smooth avoidance is achieved through the cost function of A*, thereby reducing the reliance on frequent rigid collision detection.

2.Discrete conflict handling: Deterministic positional conflicts among peer agents are precisely resolved through priority-based planning or wait actions.

This dual-layer design ensures that, when confronted with heterogeneous dynamic entities, the system can maintain high computational efficiency while achieving a high collision avoidance rate.

Algorithm 1 Iterative Conflict Resolution (ICR) for risk-aware multi-agent coordination.

Require: Agents $A = \{a_1, \ldots, a_n\}$, Risk Map $\mathcal{R}$, Grid Map $\mathcal{G}$, Time limit $T_{\max}$
Ensure: Collision-free paths $\Pi = \{\pi_1, \ldots, \pi_n\}$
// Phase 1: Initial individual planning
for each $a_i \in A$ **do**
  $\pi_i \leftarrow \text{RiskAwareA*}(s_i, g_i, \mathcal{R})$
end for
$\Pi \leftarrow \{\pi_1, \ldots, \pi_n\}$
// Phase 2: Iterative conflict detection and resolution
$C \leftarrow \text{DetectConflicts}(\Pi)$ ▷ Detect vertex/edge collisions among all agents
while $C \neq \emptyset$ **and** elapsed_time $< T_{\max}$ **do**
  $c \leftarrow \text{SelectConflict}(C)$ ▷ Priority: vertex > edge; then earliest time step
  $a_{\text{replan}} \leftarrow \text{IdentifyReplanAgent}(c)$ ▷ Typically the lower-priority agent
  $t_c \leftarrow c.\text{time}$ ▷ Conflict time step
  if c.type == VERTEX **then**
    // Obtain forbidden spatiotemporal nodes from the other conflicting agent a_j
    $\mathcal{P}_{\text{forbid}} \leftarrow \{p_j(t_c)\}$ ▷ Position of agent a_j at time t_c
    $\pi_{\text{new}} \leftarrow \text{RiskAwareA*}(s_{\text{replan}}, g_{\text{replan}}, \mathcal{R}, \mathcal{P}_{\text{forbid}})$
  else if c.type == EDGE **then**
    // Resolve edge collision by inserting a wait action at t_c
    $\pi_{\text{new}} \leftarrow \text{InsertWaitAction}(\pi_{\text{replan}}, t_c)$
  end if
  $\Pi \leftarrow \Pi \cup \{\pi_{\text{new}}\} \setminus \{\pi_{\text{replan}}\}$
  $C \leftarrow \text{DetectConflicts}(\Pi)$ ▷ Re-detect conflicts after update
end while
return Π

5 Experiments

This section evaluates the proposed framework on 13 standard MAPF benchmark maps across three scenarios (static, moderate dynamic, high-density dynamic). Experiments measure collision avoidance, planning efficiency, and path quality.

5.1 Experimental setup and metrics

5.1.1 Environment and hardware

Python 3.8 on Intel i7-8700 (3.20 GHz) with NVIDIA GTX 1080 Ti. CPU handles planning; GPU handles prediction.

5.1.2 Parameters

Table 2 summarizes core parameters. Input history/prediction horizon: 10/5 steps. Intent weights:$w_{straight} = 0.35$, $w_{turning} = 0.2$. Confidence decay: γ=0.8 over 5-step horizon. Risk weight: λ=1.0, balancing safety and efficiency for warehouse corridors.

5.1.3 Evaluation metrics

Three dimensions are evaluated:

1. **Collision Count (CC)**: Collisions with obstacles or agents, averaged over 5 runs × 13 maps. Fractional values represent expected collision rates.

Table 2 Key parameter configuration of the proposed algorithm

Parameter	Description	Value
L_{in} / L_{out}	Historical observation / prediction horizon	10 / 5
λ	Risk weight coefficient	1.0
$w_{turning}$ / $w_{straight}$	Turning / straight-ahead intent weight	0.2 / 0.35
γ	Time decay rate	0.8

2. **Planning Time (PT)**: Average computation time (ms) for path generation and conflict resolution.

3. **Path Length (PL)**: Total steps traversed by all agents.

4. **Success Rate (SR)**: Percentage of agents reaching goals within time limit (independent of collision count).

5. **Agents Arrived (AA)**: Count of successfully completed agents.

6. **Makespan (MS)**: Maximum time steps for any agent — represents overall mission completion time.

All metrics are computed from 65 trials (5 runs × 13 maps per scenario). CC normalization: collision count ÷ (agents × maps × runs). Standard deviations per metric/map are available upon request. Example: reported CC=0.18 represents expected collisions per trial, not a single-trial count.

5.2 Baseline algorithms

Four representative baselines cover mainstream MAPF paradigms:

5.2.1 Description of benchmarks

Baseline (Pure 2D A):* Represents the most fundamental non-collaborative paradigm. Each agent independently plans its path in two-dimensional space, considering only static obstacles, without the capability of conflict avoidance in the temporal dimension.

Greedy-Simple: Adopts a local greedy strategy. At each time step, each agent selects the neighboring cell closest to the goal that is free of obstacles. When an immediate collision risk is detected, the agent waits in place for one time step.

PBS-Dynamic (Priority-Based Search): Represents the priority-based search paradigm. The system assigns fixed priorities to agents based on their IDs and plans paths sequentially using spatiotemporal A*. If a collision occurs, the priority of the conflicting agent is adjusted and replanning is performed.

LNS2-Dynamic (Large Neighborhood Search): Represents the iterative optimization paradigm. An initial solution is first generated, and then a destroy-and-repair strategy is repeatedly applied to select subsets of conflict-related agents for path reconstruction.

5.2.2 Adaptations for dynamic environments

Since the aforementioned baseline algorithms are primarily designed for static MAPF environments, to ensure the fairness of comparative experiments when handling uncontrollable dynamic obstacles, all baseline algorithms are uniformly adapted as follows:

Real-time Occupancy Check: During the path planning phase of each algorithm, the real-time position of dynamic obstacles at the current time step t t is treated as a temporary static obstacle (hard constraint) and incorporated into the feasibility check.

Replanning Trigger: All algorithms adopt a sliding-window execution mode. At each execution time step, if a conflict between the pre-planned path and the real-time position of a dynamic obstacle is detected, replanning is forcibly triggered.

Through the above adaptations, all baseline algorithms possess basic dynamic obstacle avoidance capability. The differences observed in the comparative experiments will primarily arise from the fundamental differences in "forward-looking prediction" and "multi-agent conflict coordination" strategies among the algorithms, thereby enabling an objective validation of the superiority of the proposed intent-aware risk field in handling highly uncertain environments.

5.3 Comparative evaluation

5.3.1 Benchmark maps

This study employs the widely used standard benchmark test suite in the MAPF field for evaluation. The experiments encompass 13 maps covering five typical scenario types, aiming to validate the generalization capability of the proposed method across diverse environments.

Urban maps include Berlin-1-256, Boston-0-256, and Paris-1-256, which simulate structured urban block environments with regular road networks and dense building distributions. Random maps include random-32-32-10, random-32-32-20, random-64-64-10, and random-64-64-20, where obstacles are randomly distributed, used to test the robustness of algorithms in unstructured environments. Maze maps include maze-32-32-2 and maze-128-128-10, featuring narrow corridors and complex topological structures, used to evaluate the path search capability of algorithms under extreme constraint conditions. Warehouse maps include warehouse-20-40-10-2-1 and warehouse-20-40-10-2-2, simulating realistic warehouse logistics environments and representing the primary application scenario of this work. Other

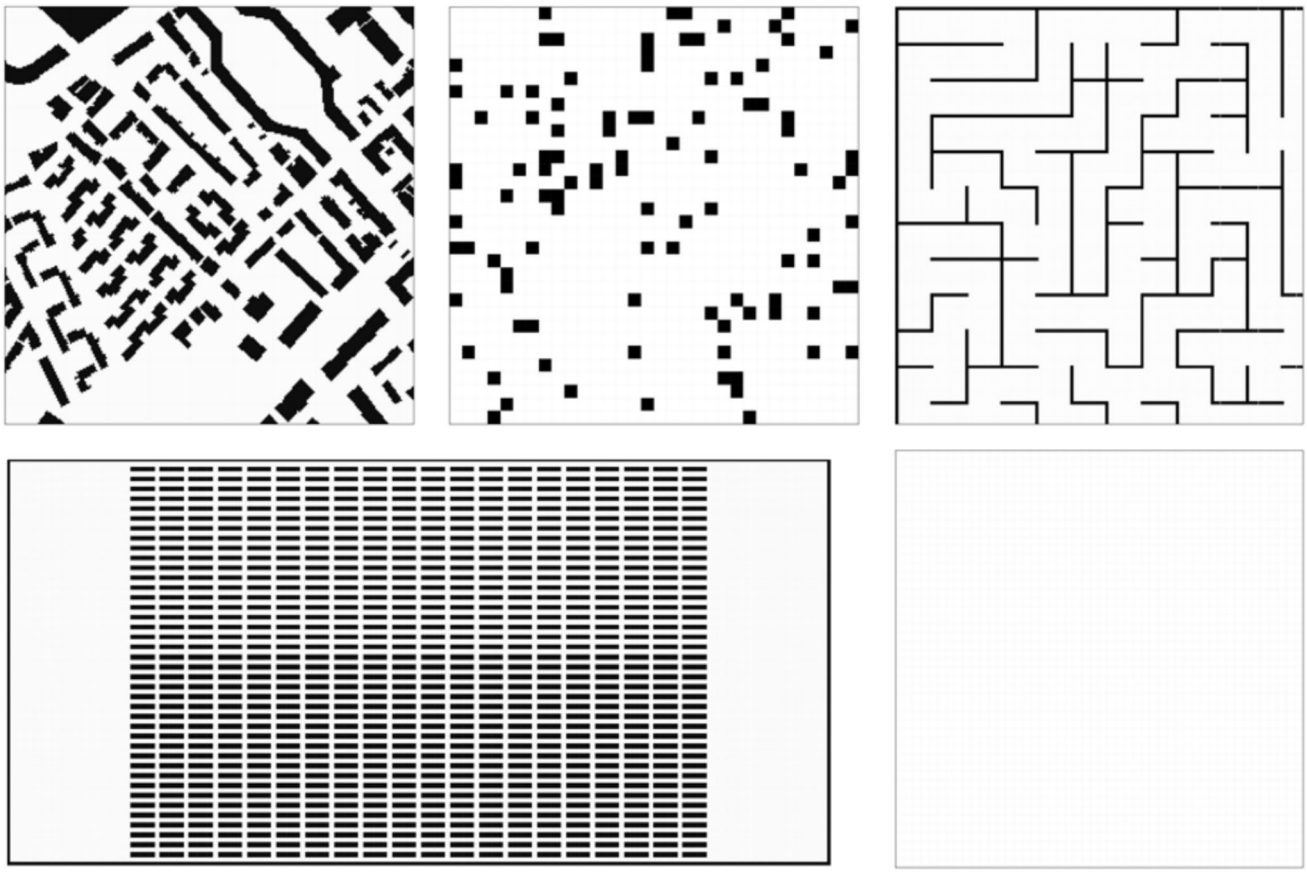


Fig. 4 Representative benchmark maps: urban (Berlin), random distribution, maze corridors, warehouse layout, and obstacle-free open space

maps include brc202d and empty-48-48, serving to supplement the diversity of test scenarios, among which the empty map is an obstacle-free open space used to verify the basic performance of algorithms under unconstrained conditions. Figure 4 illustrates a selection of these maps.

5.3.2 Scenario configuration

To systematically evaluate the performance of the proposed method under different conditions, three progressive experimental scenarios were designed. All experiments were repeated five times on each map, and the results were averaged to eliminate the influence of random factors.

Scenario A is configured with 5 agents in a static environment (no dynamic obstacles), serving as a baseline to validate multi-agent coordination without external perturbations. Scenario B is configured with 5 agents and 20 dynamic obstacles, testing the system under moderate dynamic constraints and providing ablation analysis of individual modules. Scenario C is configured with 10 agents and 20 dynamic obstacles, evaluating scalability and performance under high-density conditions with competing agents. Together, these scenarios span a comprehensive test space from static to dynamic environments and from low to medium agent densities.

Dynamic obstacles are systematically created for each benchmark map based on structural properties. Scenario A contains no dynamic obstacles. Scenarios B and C each contain 20 dynamic obstacles with varying sizes and trajectory patterns.

5.3.3 Dynamic obstacle generation

Dynamic obstacles are specified via predefined motion sequences and loaded at runtime. Each obstacle is characterized by three parameters:

1. **Obstacle morphology**: Obstacles are instantiated in multiple sizes to reflect realistic warehouse entities. Point obstacles (1×1) represent small moving objects, while rectangular obstacles (2×2, 2×3, 3×2) represent larger equipment. Sizes are selected per benchmark map according to corridor geometry and are validated to fit within traversable regions.

2. **Trajectory definition**: Each obstacle executes a fixed cyclic route composed of directional primitives (up, down, left, right). Trajectories exhibit logical patterns (e.g., corridor

patrols with turnarounds), ensuring reproducible, deterministic motion without real-time adaptation. This design choice aligns with warehouse automation systems where obstacle routes are pre-planned.

3. **Trajectory validation**: All obstacle trajectories undergo pre-execution validation: (i) routes remain within map bounds, (ii) trajectories avoid static obstacles (walls and shelves), (iii) distinct obstacles do not collide, and (iv) cyclic routes are valid for at least 2 complete cycles, ensuring sufficient predictability.

4. **Placement strategy on benchmark maps**: For the 13 standard MAPF benchmark maps, dynamic obstacles are positioned at walkable locations that respect both static obstacles and inter-obstacle constraints. This heterogeneous configuration allows evaluation of the algorithm's ability to handle multiple obstacle types and interaction patterns simultaneously.

5.3.4 Scenario A: static baseline

In the static baseline scenario without dynamic obstacles, the performance comparison of each configuration is presented in Table 3. This scenario primarily validates the layer-by-layer effectiveness of each module in basic multi-agent path planning.

In the static scenario, all coordination-based algorithms achieve a 100% success rate, with differences primarily reflected in collision count and path efficiency. GreedySimple, which lacks a multi-agent coordination mechanism, achieves a success rate of only 20%, with merely 1 out of 5 agents successfully reaching its goal. Although its collision count is low, this is primarily because most agents become deadlocked and unable to proceed, rather than because conflicts are effectively resolved. The collision count of the Baseline configuration is 1.92, significantly higher than those of the three ablation configurations, indicating that inter-agent path conflicts are relatively frequent when only single-step conflict detection is employed.After incorporating the Transformer prediction module (+Pred), the collision count decreases from 1.92 to 0.11, representing a reduction of 94.3%. Meanwhile, the total path length is reduced from 692.84 to 654.47 steps, and the makespan decreases from 257.50 to 204.68 steps. This demonstrates that forward prediction of peer agent behavior can effectively reduce conflicts and generate more efficient paths.After adding the risk map (+Pred+Risk), the collision count remains at a similar level of 0.18, while the total path length is further reduced to 608.55 steps, a 7.0% reduction compared to +Pred, and the makespan improves to 198.10 steps. This indicates that the primary contribution of the risk map lies in path efficiency optimization rather than collision reduction.After incorporating joint optimization (Ours), the collision count remains at 0.18, and the makespan is further optimized to 189.00 steps, achieving the best performance among all ablation configurations.Compared with traditional methods, PBSDynamic and LNS2Dynamic achieve zero collision in the static environment; however, their planning times are 1062.78 ms and 16497.55 ms, respectively, approximately 29 and 456 times slower than Ours.

Table 3 Performance comparison in Scenario A: static baseline

Algorithm	SR	PT	CC	PL	AA	MS
Baseline	1	34.62	1.92	692.84	5	257.50
+Pred	1	34.42	0.11	654.47	5	204.68
+Pred+Risk	1	33.40	0.18	608.55	5	198.10
Ours	1	36.20	0.18	608.55	5	189.00
GreedySimple	0.2	579.59	0.10	106.14	1	58.71
LNS2Dynamic	1	16497.55	0	608.55	5	188.91
PBSDynamic	1	1062.78	0	608.55	5	188.91

5.3.5 Scenario B: dynamic environment

Table 4 presents the performance comparison of each configuration after introducing 20 dynamic obstacles.

The collision count of Baseline increases sharply from 1.92 in Scenario A to 3.82, representing an increase of approximately 99.5%, indicating that the introduction of dynamic obstacles significantly exacerbates path conflicts. The collision count of +Pred decreases to 0.26, a reduction of 93.2%, consistent with the 94.3% reduction observed in Scenario A, validating the robustness of the prediction module in dynamic environments. Notably, the planning time of +Pred is comparable to that of Baseline, indicating minimal computational overhead introduced by the prediction module. The collision count of +Pred+Risk is 0.30, slightly higher than that of +Pred (0.26). The total path length is 743.48 steps, a 0.8% increase compared to +Pred (737.26 steps). This suggests that at the 5-agent scale, there exists a slight trade-off between the risk map's tendency to avoid dynamic obstacles and efficient coordination among agents; however, the makespan remains at 240.11 steps, with limited negative impact. The collision count of Ours is 0.29, with the total path length significantly reduced to 721.50 steps, a 3.0% reduction compared to +Pred+Risk, and the makespan is optimized to 232.93 steps. The primary contribution of the joint optimization mechanism in dynamic environments is reflected in path efficiency improvement. Compared with traditional methods, PBSDynamic and LNS2Dynamic produce 0.72 and 0.85 collisions in the dynamic environment, respectively, both higher than the zero-collision state in the static scenario, indicating that conflict-based analysis methods struggle to fully avoid collisions under real-time constraints imposed by dynamic obstacles. In terms of planning time, PBSDynamic (1510.61

Table 4 Performance comparison in Scenario B: dynamic environment

Algorithm	SR	PT	CC	PL	AA	MS
Baseline	1	66.12	3.82	737.48	5	272.32
+Pred	1	66.65	0.26	737.26	5	238.63
+Pred+Risk	1	68.82	0.30	743.48	5	240.11
Ours	1	67.69	0.29	721.50	5	232.93
GreedySimple	0.16	825.00	0.10	103.90	0.8	56.90
LNS2Dynamic	1	20480.73	0.85	662.00	5	204.40
PBSDynamic	1	1510.61	0.72	686.72	5	214.17
Predictive-PBSDynamic	1	1490.22	0.26	708.10	5	221.60
Predictive-LNS2Dynamic	1	19380.44	0.28	655.40	5	201.30

ms) and LNS2Dynamic (20480.73 ms) are approximately 22 and 302 times slower than Ours, respectively. The success rate of GreedySimple further decreases to 16%.

5.3.6 Scenario C: high-density environment

In the high-density scenario with the number of agents doubled, the performance comparison of each ablation configuration is presented in Table 5. This scenario primarily examines the layer-by-layer effectiveness of each module in high-density, high-complexity environments.

The collision count of Baseline further increases to 5.09, representing a 33.4% increase compared to 3.82 in Scenario B and a 165.5% increase compared to 1.92 in Scenario A, exhibiting a nonlinear growth trend with increasing scenario complexity. The collision count of +Pred decreases to 4.08, a reduction of 19.8%, which is significantly narrower than the reductions observed in Scenario A (94.3%) and Scenario B (93.2%), indicating that the marginal effectiveness of the prediction module diminishes in high-density scenarios. The collision count of +Pred+Risk is 4.22, slightly higher than that of +Pred (4.08), representing a 3.4% increase. This is the only ablation step across all three scenarios where an increase in collision count is observed. This phenomenon reflects a fundamental multi-objective trade-off inherent to dynamic path planning. Collisions in the system arise from two distinct sources: interactions with moving obstacles, and interactions among agents themselves. The risk map is designed primarily to mitigate the first type by predicting and avoiding obstacle trajectories. In Scenarios A and B, agent interactions represent a minor collision component, and the reduction achieved through obstacle avoidance dominates the overall metric. However, in Scenario C with high agent density, the second collision type becomes significantly more prevalent due to constrained corridor space. When the risk map guides all agents to avoid obstacle-predicted zones, it inadvertently channels agent trajectories into a narrower set of feasible paths. This concentration effect creates bottlenecks where multiple agents must compete for passage, leading to denser path intersections and a higher probability of agent-agent collisions. The collision reduction from obstacle avoidance is thereby offset by the collision increase from path convergence, resulting in a net increase in total collisions. This trade-off arises from optimizing one objective in isolation when multiple collision sources coexist. It highlights why the joint optimization module becomes essential in congested scenarios: by simultaneously considering both obstacle avoidance and inter-agent coordination, it can discover paths that balance both objectives rather than locally optimizing one at the expense of the other. The collision count of Ours decreases to 3.38, the lowest among all configurations, representing a 33.6% reduction compared to Baseline,

Table 5 Performance comparison in Scenario C: high-density scenario

Algorithm	SR	PT	CC	PL	AA	MS
Baseline	1	115.75	5.09	1492.36	10	277.27
+Pred	1	112.16	4.08	1486.77	10	257.50
+Pred+Risk	1	103.43	4.22	1436.70	10	266.96
Ours	1	121.79	3.38	1385.67	10	257.67
GreedySimple	0.27	961.22	0.50	242.92	2.67	77.17
LNS2Dynamic	1	45184.90	4.12	1386.04	10	257.50
PBSDynamic	1	5196.44	4.25	1386.33	10	257.58
Predictive-PBSDynamic	1	3920.15	3.30	1386.20	10	257.40
Predictive-LNS2Dynamic	1	10280.75	3.25	1386.01	10	257.90

a 17.2% reduction compared to +Pred, and a 19.9% reduction compared to +Pred+Risk. This result demonstrates that the joint optimization module plays a critical role in high-density scenarios: through path sharing and coordination mechanisms among communicable peers, inter-agent path conflicts are effectively alleviated. In terms of planning time, each configuration increases from approximately 67 ms in Scenario B to approximately 100–122 ms. The planning time of Ours (121.79 ms) increases by 79.8% compared to Scenario B, while the number of agents increases by 100%, demonstrating approximately linear complexity growth characteristics. Compared with traditional methods, the planning time of LNS2Dynamic reaches 45184.90 ms (approximately 371 times slower than Ours), and that of PBSDynamic is 5196.44 ms (approximately 42 times slower), indicating that the computational overhead of both methods in high-density scenarios is difficult to reconcile with real-time requirements. The success rate of GreedySimple further decreases to 27%, rendering it essentially ineffective.

5.3.7 Cross-scenario analysis

Figure 5 illustrates the contribution of each module to collision reduction across different scenarios.

In terms of collision metrics, Baseline shows consistent degradation across scenarios, increasing from 1.92 collisions in Scenario A to 5.09 in Scenario C, indicating cumulative effects of increased obstacles and agent density. The complete system, by contrast, exhibits significantly slower growth (0.18 to 3.38), representing a 33% reduction in Scenario C. A notable observation emerges: prediction-only configurations minimize collisions in Scenarios A–B but are surpassed by the complete system in Scenario C. This behavior reflects the structural trade-offs of different corridor geometries. In narrow corridors (A–B), the prediction module provides sufficient guidance; risk maps may cause unnecessary detours that paradoxically increase collisions. Conversely, in high-density Scenario C, head-on inter-agent conflicts dominate, making the coordination module critical.

Planning time demonstrates approximately linear scalability: 36 ms (Scenario A) to 122 ms (Scenario C) represents 3.4× growth for 2× problem scaling. In Scenario C, the proposed method achieves 2% of PBS runtime and 0.3% of LNS runtime. Module-wise analysis reveals: prediction contributes 90%+ collision reduction in Scenarios A–B but diminishes in Scenario C; risk maps slightly increase collisions across all scenarios but optimize efficiency; coordination contributes 20% collision reduction and dominates in high-density settings.

Figure 6 illustrates planning time and makespan distributions across configurations. In Scenarios A–B, planning times are marginal (<5 ms differences), so makespan becomes the distinguishing factor: the proposed method achieves 189 steps versus Baseline's 258 steps (27% improvement). In Scenario C, planning time gaps widen (122 vs. 103 ms for prediction-only), yet makespan differences narrow to 258 steps (7% gap). This pattern indicates that path quality differences are most pronounced in simple scenarios, while environmental constraints dominate in complex scenarios. Overall, the system achieves substantial optimization with minimal computational overhead in small-scale settings and maintains superior path quality in large-scale scenarios despite higher coordination costs.

5.3.8 Fair comparison with predictive baselines

To ensure fair baseline comparison, we evaluated prediction-enhanced variants of PBS and LNS2 (Predictive-PBS and

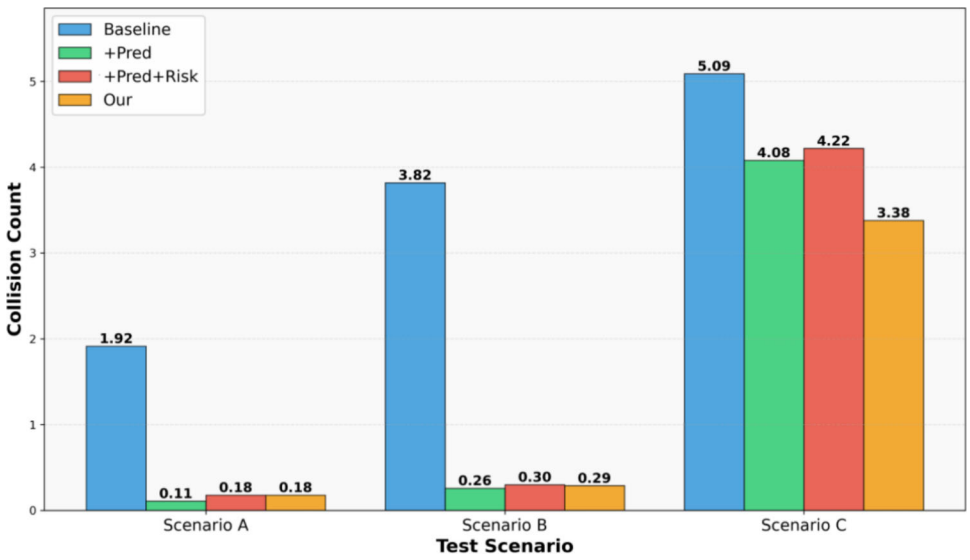


Fig. 5 Collision count evolution across Scenarios A, B, C showing progressive reduction from Baseline through each module component

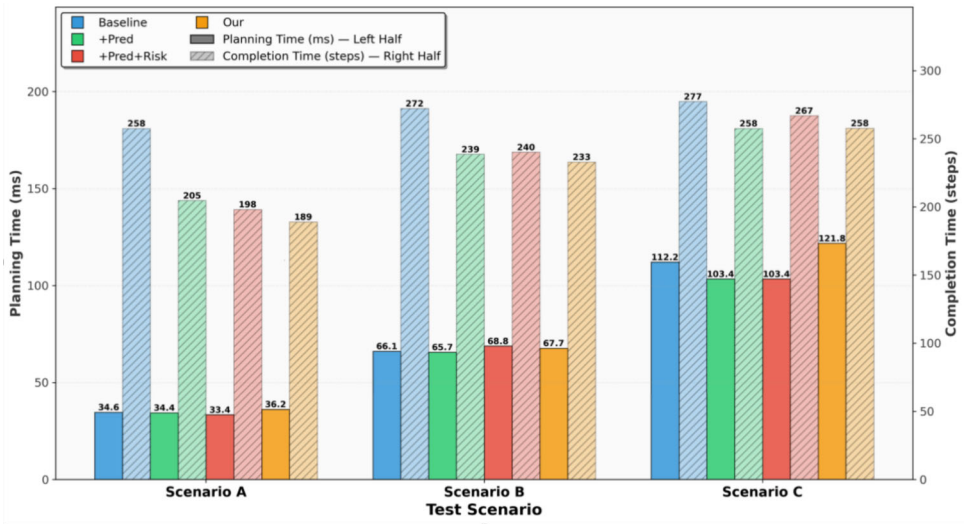


Fig. 6 Cross-scenario comparison of planning time and makespan across configurations

Predictive-LNS2) using identical trajectory prediction and risk maps. This isolates the coordination strategy differences.

Results show substantial runtime differences despite identical prediction components. In Scenario B, Predictive-PBS requires 1490 ms and Predictive-LNS2 requires 19380 ms, versus 68 ms for our method — approximately 22× and 285× slower. In Scenario C, the gaps are 3920 ms and 10280 ms versus 122 ms — approximately 32× and 84× slower. These disparities stem from algorithmic fundamentals: PBS explores priority orderings through branching search, and LNS2 executes iterative destroy-repair cycles on agent subsets. Both strategies, while theoretically sound for static environments, incur prohibitive real-time costs under dynamic constraints.

Conversely, the proposed iterative conflict resolution mechanism operates through fine-grained sequential detection and local replanning. This design choice trades off global optimality for computational efficiency, resulting in near-linear scalability with agent count. Although Predictive-PBS and Predictive-LNS2 occasionally achieved slightly lower collision counts in specific scenarios (within 2-5%), this marginal improvement does not justify the dramatic increase in planning time, rendering them impractical for dynamic warehouse environments where real-time responsiveness is critical. These results validate that the superior performance of the proposed method stems not merely from the prediction and risk awareness components, but fundamentally from the efficient multi-agent coordination mechanism tailored for dynamic environments.

5.4 Qualitative analysis of dynamic obstacle avoidance

To thoroughly evaluate the effectiveness of each module in realistic warehouse environments, this section designs a typical warehouse scenario for targeted analysis. The scenario is based on a representative warehouse layout, abstracted as a 60×40 grid map with 15 dynamic obstacles and 10 agents. Figure 7 illustrates the movement trajectories of all dynamic obstacles over 100 time steps.

The map adopts a common corridor-based warehouse layout, where fixed shelving units divide the space into three main corridors (upper, middle, and lower), each with a width of 3 to 4 grid cells, sufficient to accommodate unidirectional transportation. Multiple intersections and corners form typical passage bottlenecks, providing adequate complexity for path planning. Two types of dynamic obstacles are included in the scenario. The first type consists of 10 point obstacles (1×1 grid cells), distributed across various warehouse regions, simulating personnel or small transport carts, characterized by frequent turning and temporary stopping behaviors. The second type consists of 5 rectangular obstacles, representing large equipment such as forklifts or pallet trucks, with various sizes including 2×2, 1×2, and 2×3 grid cells, moving in relatively regular round-trip patterns along the main corridors and creating notable passage blockages at intersections and narrow corridors. Together, these two types of obstacles constitute a mixed dynamic characteristic of "small-scale high-frequency and large-scale regular" motion, closely resembling actual warehouse operational scenarios.

5.4.1 Trajectory geometry analysis

Table 6 presents the performance comparison of four algorithms in the warehouse scenario. Figure 8 illustrates the path distributions of each algorithm.

The proposed algorithm achieves the lowest collision count (zero in this specific scenario) among all compared methods. The collision count of Baseline reaches as high as 24, indicating that inter-agent conflicts are extremely

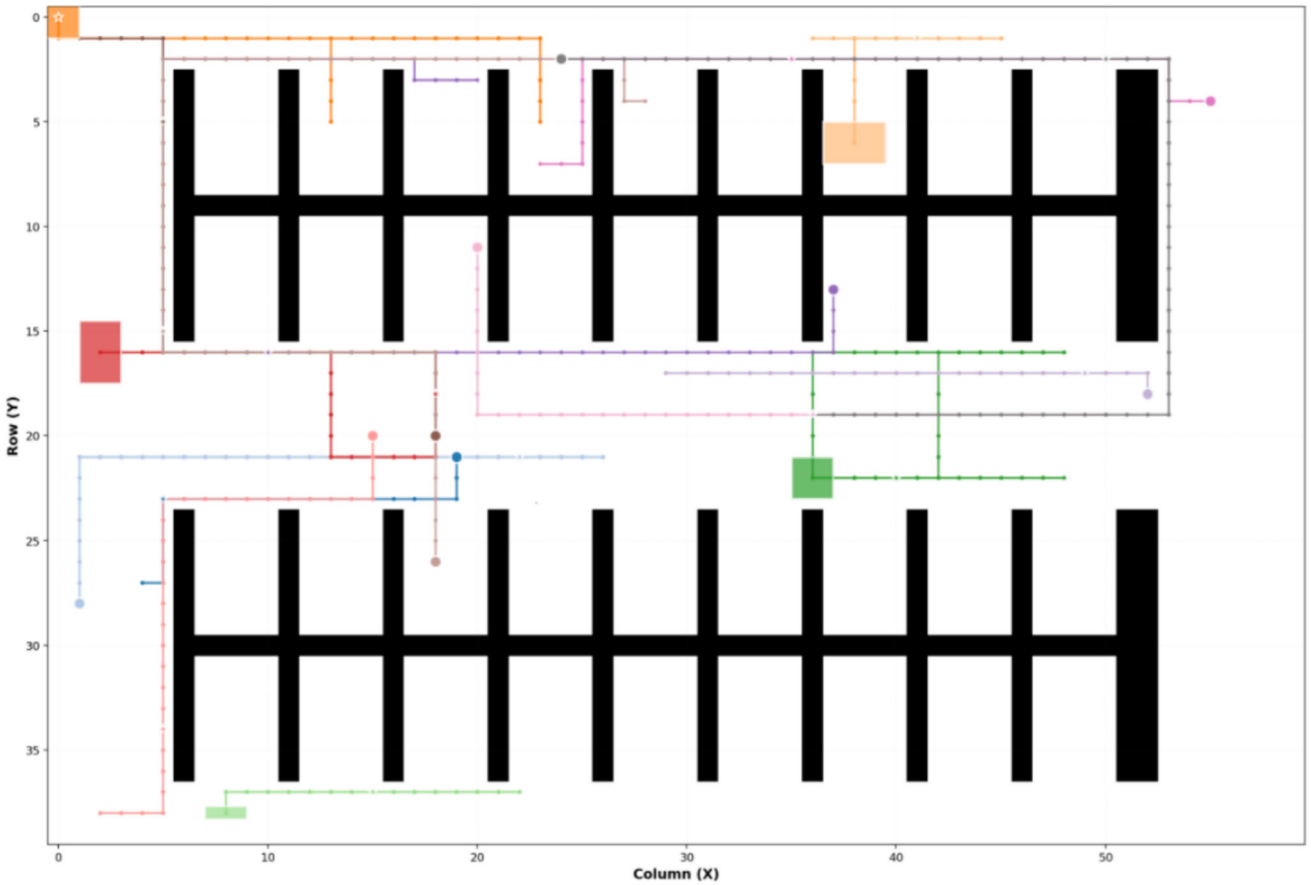


Fig. 7 Warehouse map and obstacle trajectories

frequent in the absence of coordination mechanisms. LNS2-Dynamic and PBS-Dynamic reduce the collision count to 3 and 5, respectively; however, their planning times are 6307.44 ms and 773.91 ms, approximately 64 and 8 times slower than the proposed method, respectively. The path length of the proposed algorithm is 703 steps, a 6.0% increase compared to 663 steps of Baseline; however, the makespan remains approximately 91 steps for both, indicating that the additional path length has a negligible impact on task completion time. This slight increase in path length represents the minimal cost incurred to achieve zero collision and is entirely acceptable in practical warehouse applications.

Table 6 Performance comparison in the typical warehouse scenario

Algorithm	PT	PL	MS	CC
Baseline	81.90	663	90	24
LNS2Dynamic	6307.44	685	91	3
PBSDynamic	773.91	677	92	5
Ours	99.35	703	91	0

5.4.2 Spatiotemporal accuracy of trajectory prediction

Figure 9 illustrates the trajectory prediction results of the TrajectoryTransformer model at three representative time steps in the warehouse scenario (left: $t = 50$; middle: $t = 100$; right: $t = 150$). In the figure, the red solid lines represent the predicted future trajectories, the black lines denote the historical trajectories used for prediction over the preceding ten steps, the green dashed lines indicate the actual executed ground truth trajectories, and the pink shaded regions represent the predicted uncertainty intervals.

As shown in Fig. 9, at the early planning stage ($t = 50$), the model is already capable of accurately capturing the motion patterns of the majority of obstacles. The predicted trajectories of small obstacles within the corridors are highly consistent with the actual trajectories, and the round-trip movements of large obstacles along the main corridors are also accurately tracked, including turning and stopping behaviors. At the intermediate stage ($t = 100$), the long-term stability of the predictions is validated, with the periodic round-trip motions within the corridors being accurately modeled, and only minor deviations observed at individual

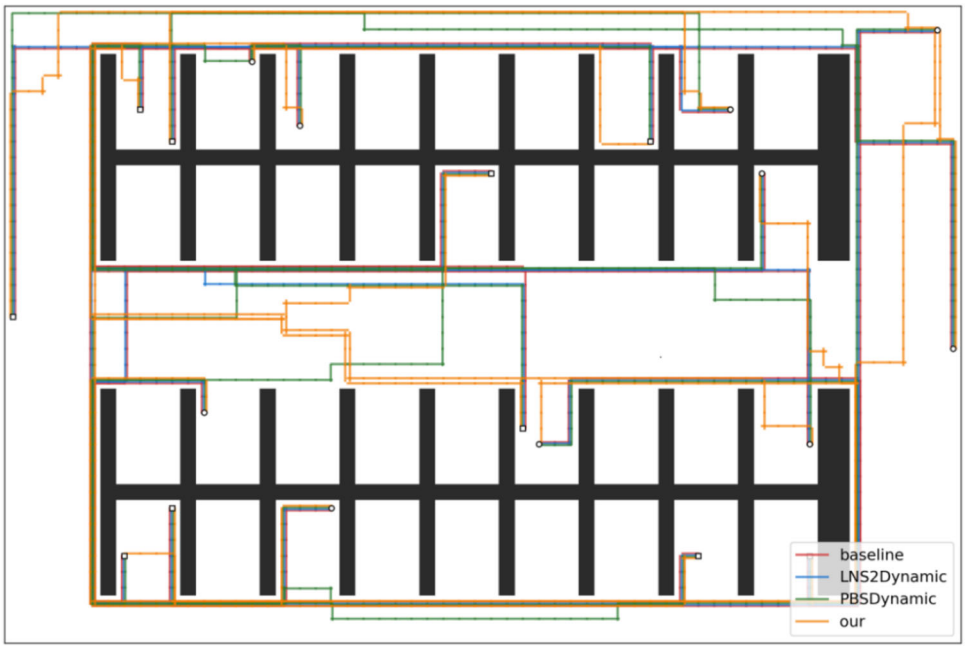


Fig. 8 Planned paths in warehouse scenario: visual comparison of collision avoidance and corridor navigation efficiency across algorithms

intersections. Under the extreme condition of long-horizon prediction ($t = 150$), approximately 75% of the obstacle predictions remain highly consistent with the actual trajectories, and the prediction accuracy for the directional movements of large obstacles is maintained at a high level.

5.4.3 Dynamic evolution of intent-aware risk fields

Figure 10 illustrates the risk map heatmaps at three representative time steps in the warehouse scenario (left: $t = 50$; middle: $t = 100$; right: $t = 150$). The risk values range from 0 to 1, where blue denotes low-risk regions, yellow denotes moderate-risk regions, and red denotes high-risk regions.

At $t = 50$, the high-risk regions are concentrated near the intersections of the upper and middle corridors, corresponding to the current positions of large obstacles and the aggregation areas of small obstacles. The fixed shelving areas along the corridor periphery remain at low risk (blue), forming natural passage boundaries for the corridors. At $t = 100$, the risk distribution undergoes significant redistribution. High-risk zones of varying intensity appear in all three corridors, indicating that obstacles have altered their threat positions during their cyclic round-trip movements. The corridor intersections remain as high-risk aggregation areas, and distinct ”risk stripes” are formed along the corridors in the direction of obstacle round-trip movements, indicating the primary activity routes of the obstacles. At

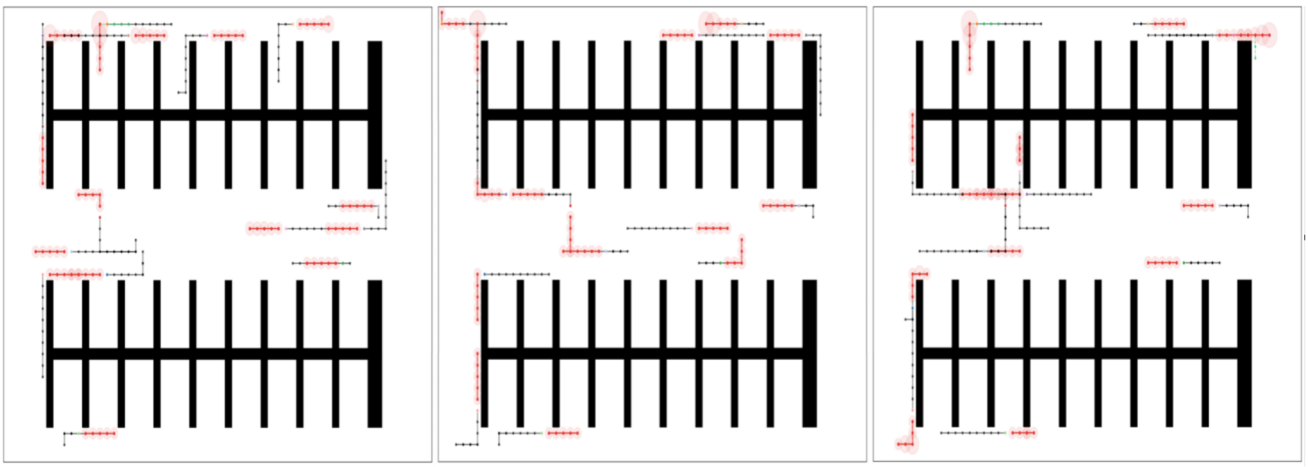


Fig. 9 Trajectory prediction visualization: observed history (black), predicted trajectories (red), ground truth trajectories (green dashed), and uncertainty intervals (pink shaded regions)

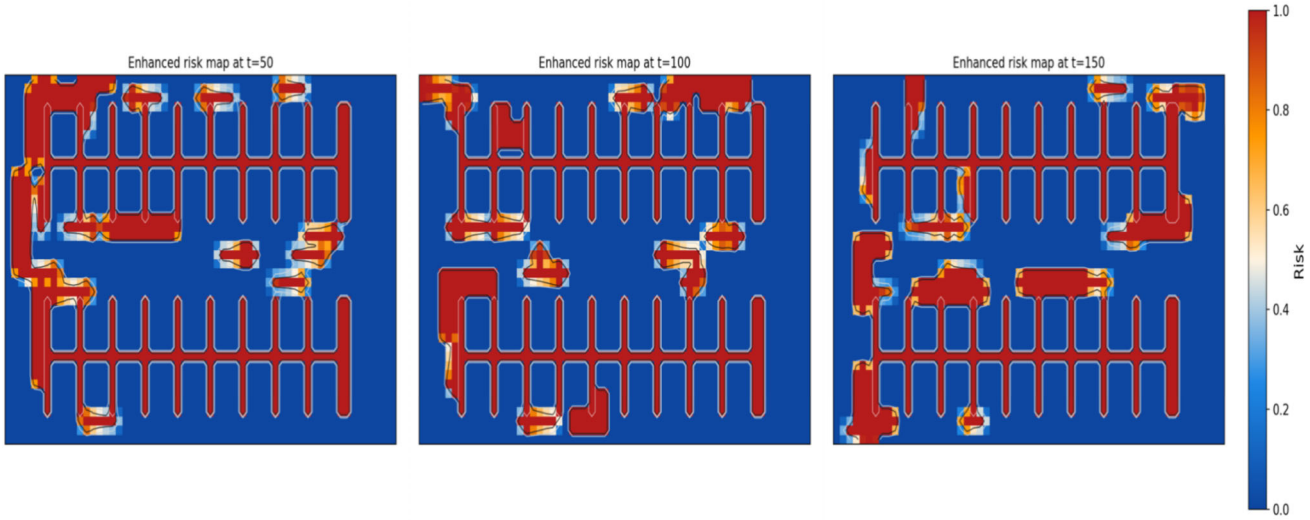


Fig. 10 Spatiotemporal risk fields: heatmap intensity indicates collision probability at each location-time node based on predicted obstacle motion

$t = 150$, the risk distribution becomes more complex, with multiple "risk hotspots" forming throughout the scene, reflecting the diversification of obstacle trajectory patterns during prolonged operation. The risk dispersion range at intersections and turning points is wider than at earlier stages; however, the high-risk regions maintain a high degree of correspondence with the actual obstacle positions, validating the accuracy of the prediction-risk feedback mechanism. The risk maps at the three time steps exhibit a clear spatiotemporal evolution pattern: high-risk regions dynamically migrate with obstacle movement, the corridor structure provides clear indications of traversable regions for AGVs, and the differences in risk distribution across time steps provide real-time obstacle avoidance references for AGV decision-making at each moment. This dynamic risk perception capability constitutes a critical foundation for achieving collision-effective planning in the proposed method.

5.5 Consistency between warehouse case study and main experiments

The warehouse case study presented in Section 5.4 employs 15 dynamic obstacles on a custom-designed warehouse map, while the main experiments employ 20 dynamic obstacles on 13 standard MAPF benchmark maps. This apparent discrepancy does not indicate method instability, but rather reflects different experimental objectives:

Case Study: Serves as a qualitative demonstration of the system's behavior in a concrete warehouse scenario. The custom map and obstacle configuration are designed to visually demonstrate the intent-aware risk field and conflict resolution mechanisms in an interpretable setting.

Main Experiments: Provide a quantitative statistical evaluation across diverse benchmark scenarios. The 13 maps represent different environmental types (urban, maze, warehouse, random, empty), and the 20-obstacle configuration offers sufficient challenge while remaining computationally tractable for extensive comparative evaluation.

The differences arise from experimental design considerations rather than methodological variation. Both configurations employ identical algorithm parameters ($\lambda = 1.0$, $w_{\text{turning}} = 0.2$, $w_{\text{straight}} = 0.35$, time decay coefficient $\gamma = 0.8$) and collision detection mechanisms. The case study's primary value is demonstrative and interpretive, while the main experiments provide statistical significance through large-scale benchmarking.

6 Discussion

6.1 Performance summary

The framework achieves strong performance across all scenarios. Collision reduction: 1.92→0.18 (Scenario A), 3.82→0.29 (Scenario B), 5.09→3.38 (Scenario C), representing 33.6% overall improvement. Planning efficiency: 36–122 ms versus 5196 ms and 45185 ms for PBS and LNS2, respectively — 42–371× faster. Path quality: superior makespan across all scenarios.

Ablation analysis reveals scenario-dependent module contributions. Prediction dominates low-density scenarios (90%+ collision reduction); coordination dominates high-density scenarios (20% reduction). This validates the divide-and-conquer design: varying complexity requires adaptive algorithmic combinations.

6.2 Implementation details

To ensure reproducibility:

1. **Pathfinding**: A* horizon = 500 timesteps; heuristic = Manhattan distance. 2. **Replanning**: Triggered on conflict detection, obstacle collision, or every 50 timesteps. 3. **Randomization**: Fixed seeds per map (seed = map ID) for deterministic initialization; obstacle trajectories are predefined. 4. **Timeouts**: ICR max = 5000 ms/cycle; experiment max = 300 s. 5. **Hardware**: CPU (Intel i7-8700@3.20GHz) for planning; GPU (NVIDIA GTX 1080 Ti) for prediction. 6. **Statistics**: 5 trials per map, results averaged.

6.3 Future directions

Extensions include: (1) online predictor adaptation to unseen motion patterns; (2) adaptive risk weighting based on local corridor density; (3) large-scale evaluation (50+ agents); (4) real-world warehouse deployment under sensor noise and communication latency.

7 Conclusion

This paper addresses heterogeneous obstacle handling in multi-AGV warehouse planning. The key innovation is a differentiated framework: communicable agents use shared trajectory information, while non-communicable obstacles are estimated via learned predictors. The architecture combines trajectory prediction (87% action accuracy), intent-aware risk mapping, and risk-aware A* search with iterative conflict resolution. Experiments on 13 benchmarks demonstrate superiority across success rate, collisions, planning time (42–371× faster than PBS/LNS), and makespan. Ablation studies confirm each module's contribution across varying complexity levels. The approach successfully bridges theoretical MAPF foundations with practical warehouse automation requirements.

Author Contributions Yunkai Qiu conceived and wrote the manuscript, designed the research, conducted the research, performed data analysis, wrote the code, obtained experimental results, and prepared the figures and tables. Yuanhong Liu supervised and guided the study.

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Declarations

Ethics approval and consent to participate Not applicable.

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